

LearnWave — Web App UX & Feature Spec

Goal: Build a stunning, modern, animated platform inspired by reference images, with separate portals for Admin, Mentor, and Student. Only Student and Mentor can sign up; Admin is login-only. Mentors require resume upload + admin approval before access. Admin has full privileges (uploading, approving, analytics, moderation, configuration).

1) Visual Direction & Motion Language

- Style: Clean, high-contrast, glassmorphism accents, soft gradients, neumorphic cards in selective areas.
- 3D & Backgrounds: Three.js + react-three-fiber for moving backgrounds; shader-based animated gradients.
- Motion: Framer Motion, GSAP, Lottie.
- Hover states: Magnetic buttons, tilt-on-hover, elevated shadows.
- Responsiveness: Fluid grids and mobile-first layouts.

2) Open-Source UI & Dev Stack

- Next.js (App Router), Tailwind, shadcn/ui, lucide-react.
- Animations: Framer Motion, GSAP, Lottie.
- 3D: three, @react-three/fiber.
- State: Zustand + TanStack Query.
- Forms: react-hook-form + zod.
- Auth: NextAuth.js, RBAC.
- Backend: Node/NestJS or Express; Prisma + PostgreSQL.
- Search: Meilisearch.
- Realtime: Pusher/Socket.io.
- AI: provider-agnostic.
- Storage: S3-compatible.

3) Information Architecture & Routing

- Public: /, /about, /auth/*
- Student: /s/*
- Mentor: /m/*
- Admin: /admin/*

4) Role Matrix

Students, Mentors (pending/approved), and Admin with detailed capability segregation including resource creation, approvals, moderation, analytics, and announcements.

5) Authentication & Onboarding

Students: email/password + USN/branch/semester.

Mentors: email/password + resume + admin gating.

Admin: login only.

6) Admin Dashboard — Powerful Controls

Mentor approvals, resource management, user CRUD, forum moderation, announcements, timetable uploads, placements, payouts, detailed analytics, audit logs, feature flags.

7) Student Experience

Dashboard, resources, AI tools, forum, timetable, placements, results, notifications.

8) Mentor Experience

Pending: checklist; Approved: content studio, sessions, payouts.

9) Data Model (PostgreSQL)

users, students, mentors, resources, announcements, timetables, forum_posts, bookings, notifications, audit_logs.

10) API Surface

Auth, mentor approvals, resources, search, forum, timetable, results, AI, notifications.

11) Security & Privacy

RBAC, validation, file scanning, rate limiting, PII minimization, encrypted fields, audit trails.

12) Performance & Quality

Lighthouse ≥90, 60fps motion, code-splitting, optimized images. Tests with Vitest, Playwright, k6.

13) Accessibility & i18n

WCAG 2.2 AA, ARIA roles, keyboard-first, multiple locales.

14) Deployment & Ops

Multiple environments, monitoring, backups, CI/CD.

15) Roadmap (6–8 Weeks)

Week-by-week breakdown from auth to dashboards to AI to analytics.

16) Acceptance Criteria

Stable RBAC, no console errors, searchable resources, functional approvals, functional booking.

17) Microinteractions

Animated buttons, parallax cards, sliding tabs, toasts, playful empty states.

18) Nice-to-Haves

Gamification, PWA, A/B experiments.

Build Notes

Ensure animations are interruptible, low-power mode toggle, open-source compliance.